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THEME–A

India and the World, Land and the People

India is a land of great geographical and cultural diversity. It has majestic mountains, fertile plains, vast deserts, plateaus, long coastlines, and beautiful islands. These physical features influence the climate, vegetation, wildlife, occupations, and lifestyles of the people. India is home to people of different languages, religions, traditions, and cultures who live together with mutual respect. Its location in South Asia has connected it with many countries through trade, travel, and cultural exchange since ancient times. Understanding India's land and people helps us appreciate its rich heritage, natural resources, unity in diversity, and important role in building peaceful and friendly relations with the world.

1 Geographical Diversity of India

Let's Learn About

- The main physical regions of India.
- The important landforms such as mountains, plateaus, plains, deserts and coasts.
- The special features of each region and how they influence climate, culture and daily life.
- How people live and adjust to different landforms and climates across India?

Let's Gear Up

Vanya and Kabir's India Journey

"Look at this, Kabir." Vanya said, pointing to a map on the wall. "Mahatma Gandhi once said India's land makes it very special. What do you think he meant?"

Kabir opened his atlas and smiled. "It means India has everything! Snowy mountains in the north, deserts in the west, green plains, forests, long rivers and islands in the seas!"

Vanya's eyes sparkled. "So our country is like one big home with many different rooms for everyone?"

"Exactly!" Kabir replied. "Remember when astronaut Kalpana Chawla looked at India from space? She said it looked so beautiful, like a jewel on Earth."

"Wow, that's amazing!" Vanya exclaimed. "I want to explore everything – the mountains, plains, deserts, plateaus, coasts and islands."

"Let's do it" Kabir said, spreading the map on the table. "Our adventure across India starts now – from the peaks to the shores. Ready?"

"Ready!" Vanya grinned. "Let's go on our journey!"

INTRODUCTION

India is the seventh-largest country in the world and is located in South Asia. Along with its its neighbours—Pakistan, China, Nepal, Bhutan, Bangladesh, Myanmar, Afghanistan, the Maldives and Sri Lanka—it forms a special region known as the Indian Subcontinent. To understand India's large and varied landscape, we divide it into five main physical regions:

- The Himalayan Mountains
- The Indo-Gangetic Plains
- The Thar Desert
- The Peninsular Plateau
- The Coastal Plains and Islands

Each of these regions is different in height, climate, natural plants, and the way people use the land.

The Himalayas form India's northern boundary and stand like a strong natural wall. In the west, the Thar Desert and the Arabian Sea create clear physical borders. The Indian Ocean lies to the south, while the Bay of Bengal is in the east. These natural features have greatly influenced India's weather, history and way of life.

In-Text Questions

1. India is the _____ largest country in the world.
2. The _____ Mountains form the northern boundary of India.

THE HIMALAYAS

The Himalayas are the highest mountain range in the world. They stretch across many countries and stay covered with snow for most of the year.

During summer, the melting snow gives rise to great rivers like the Ganga, Indus, and Brahmaputra. These rivers provide water for farming, drinking and industries and support the lives of millions of people. That is why the Himalayas are known as the 'Water Tower of Asia.'

The Himalayas are also holy for many people. Many temples and monasteries are located in these mountains. Monks and pilgrims from different parts of the world visit these sacred places to pray, think peacefully and meditate.

The Himalayan mountain range spreads across five countries—India, Nepal, Bhutan, China, and Pakistan. It is home to the famous "Eight Thousanders," which are very high peaks rising above 8,000 metres.

How the Himalayas Were Formed – A Fascinating Story

The Himalayas, known as the 'abode of snow,' have a wonderful story that began millions of years ago. Long ago, India was part of a huge landmass called Gondwana, which was located near Africa. Slowly, the Indian landmass started moving northward and finally crashed into the large Eurasian Plate. This powerful collision made the Earth's crust bend and rise, forming the mighty Himalayas.

This movement is still going on today. The Indian Plate continues to move north at about 5 centimetres every year. This means the Himalayas are still growing. In fact, these mountains rise nearly 5 millimetres each year, which becomes about 5 metres in a thousand years.

This constant movement not only shapes the land but also affects the climate and environment of the region. The Himalayas control weather patterns and are home to many plants and animals, making them very important for nature.

Formation of the Himalayas

The word 'Himalaya' comes from two Sanskrit words—'hima' meaning snow and 'alaya' meaning home—so it means 'home of snow.' The Himalayas are divided into three main parallel ranges: Himadri, Himachal, and the Shivalik Hills, and each has its own special features.

The Himadri (Greater Himalayas)

The Himadri, also called the Greater Himalayas, is the highest and roughest mountain range. It has some of the tallest peaks in the world, such as Mount Everest and Kanchenjunga. This range stays covered with snow all year round. Very few people live here because the climate is extremely cold and the height is very great. Just below the Himadri lies the Himachal range.

The Himachal (Middle or Lesser Himalayas)

The Himachal, also called the Middle or Lesser Himalayas, has a pleasant and moderate climate. It is rich in plants and animals. This region is famous for beautiful hill stations like Nainital, Shimla, Darjeeling, and Mussoorie. Because of its cool weather and scenic beauty, it is a popular place for tourists and settlements.

The Shivalik Hills (Outer Himalayas)

The Shivalik Hills, also called the Outer Himalayas, are the lowest range and form the foothills of the Himalayas. They have soft, rolling hills, thick forests and many wild animals. The Shivaliks act as a natural boundary between the high mountains and the northern plains.

Life and Nature in the Himalayas

In the western Himalayas, people have created a unique way of building houses known as kath-kuni or dhajji dewari. These houses are made by using local stone and wood together. This method keeps the homes warm during the cold winters and also helps them stay strong during small earthquakes. It shows how people wisely adjust their houses to live comfortably in the harsh mountain climate.

The Himalayas are also full of rich plant and animal life. The Great Himalayan National Park in Himachal Pradesh protects many rare species of plants and animals. It has been recognized as a World Heritage Site by UNESCO. The park is protected with the help of both the government and nearby village communities, who work together to take care of this valuable natural heritage.

Do you know?

The Himalayan monal, a brightly coloured bird, is the state bird of Himachal Pradesh and among the many lovely species living in the Great Himalayan National Park.

This region is well known for animals like the snow leopard, the bright Himalayan monal bird, and lovely rhododendron flowers. These flowers are often used to prepare tasty sherbet. People in the mountains also make beautiful handicrafts from wool, wood and natural fibres, showing their colourful culture and traditions of Himalayan life.

Great Himalayan National Park, Conservation Area

Himalayan monal (male)

Rhododendron

In-Text Questions

1. Which plate did the Indian landmass collide with to form the Himalayas?
2. Name one traditional building method used in the western Himalayas to make houses earthquake-resistant.
3. Which UNESCO World Heritage Site in Himachal Pradesh is home to many plants and animals?
4. What do the Sanskrit words hima and alaya mean when combined to form 'Himalaya'?

THE COLD DESERT: LADAKH

When we think of a desert, we usually imagine hot and dry land. But there are also cold deserts and Ladakh in India is one of them. Ladakh is very cold, with winter temperatures dropping below -30°C. It receives very little rainfall and the land is rocky and uneven, with deep valleys and lakes such as Pangong Tso (tso means lake in the local language).

Moonland, Ladakh

The land in Ladakh looks like the surface of the moon, so it is sometimes called "moonland." Geologists say this special landscape was formed when the Indian Plate crashed into the Eurasian Plate, pushing the mountains up from what used to be part of an ocean.

Pangong Tso, Ladakh

Most of the land has sand and clay. Over many years, wind and rain have shaped the mountains into unusual and beautiful forms.

Analyze Thoughts

Pick a place with extreme weather, like the Thar Desert. Learn how people survive there—their food, clothes, and homes. Then compare it with the way people live in Ladakh.

People and Culture of Ladakh

Even with the harsh weather, Ladakh is home to animals such as the snow leopard, ibex, and Tibetan antelope. The people live simple lives and enjoy bright festivals like Losar and the Hemis Festival. The region is also famous for its ancient monasteries.

Monastery

Losar Festival

Himalayan Ibex

Yaks used for transport

THE GANGETIC PLAINS

Stretching south from the foothills of the Himalayas, the Gangetic Plains form a wide, low-lying region full of fertile soil. Rivers like the Ganga, Yamuna and Brahmaputra bring mineral-rich silt,

making the land perfect for farming. These rivers are important for producing hydroelectricity, transport and trade. Roads and railways help people and goods travel long distances easily. For thousands of years, rivers such as the Ganga and Brahmaputra have been used for travel and trade. A large part of India's population lives in these plains.

The Indian Gharial

Modern Farming

Multiple Cropping

Do you know?

Most rivers in India are named after goddesses, like Ganga, Yamuna, and Kaveri. But the Brahmaputra is different. Its name means "son of Brahma." In summer, its water level rises because of melting snow.

THE GREAT INDIAN DESERT OR THAR DESERT

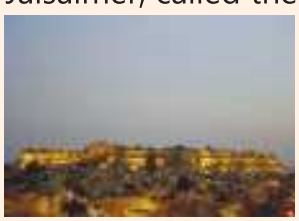
The Thar Desert is a huge area of golden sand dunes, rough land and endless sky.

Sand dunes are made when strong winds move the sand into hill-like shapes, some reaching up to 150 metres high.

Rajasthan Thar Dessert

The Thar is a large dry region. Most of it is in India, covering Rajasthan, Gujarat, Punjab and Haryana. The desert also acts as a natural barrier because its tough conditions make it hard for people and animals to move around.

Jaisalmer, called the "Golden City," is in the heart of the Thar Desert in India. The Jaisalmer Fort here is a UNESCO World Heritage Site.



Jaisalmer Fort

Rajasthan is also known for its traditional water-saving methods. Taanka or Kunds are used to collect and store rainwater for drinking during dry seasons.

Taanka

THE ARAVALLI HILLS

The Aravalli Hills are one of the oldest mountain ranges in the world, around 2.5 billion years old. The range is made up of many low hills and ridges. The highest peak, Guru Shikhar on Mount Abu, is over 1,700 metres high, while most hills are between 300 and 900 metres. Interestingly, in just four and a half hours by car, you can travel from the cool hills of Mount Abu to the hot sands of Jodhpur in the Thar Desert, showing how different the landscapes are.

The Aravalli are very important for north-western India's geography and climate. They act as a natural barrier, helping to stop the Thar Desert from spreading eastward. These hills are also rich in minerals like marble, granite, zinc, and copper. Mining and construction have been carried out here for many years. At Zawar, people learned how to extract zinc from rocks over 800 years ago,

making it one of the earliest zinc production centers in the world. Many famous forts, including Chittorgarh, Kumbhalgarh, and Ranthambore, are also found in the Aravallis.

Kumbhalgarh Fort

Chittorgarh Fort

Ranthambore Fort

THE PENINSULAR PLATEAU

The Peninsular Plateau is one of India's oldest and most important landforms, covering a large part of central and southern India. It is triangular in shape and is called a peninsula because it is surrounded by water on three sides—the Arabian Sea, the Bay of Bengal, and the Indian Ocean.

The plateau is bordered by two main mountain ranges:

The Western Ghats on the west, which are taller and form a continuous wall along the western coast. During the monsoon, many waterfalls appear along their steep slopes. These hills are a UNESCO World Heritage Site, rich in plants and animals, and the source of many rivers. The northern part of this range is called the Sahyadri Hills.

Power House Waterfalls at Periyakanal

The Eastern Ghats on the east, which are lower and broken into smaller hills stretching along the eastern coast. Between these ranges lies the Deccan Plateau, a large flat area. Several rivers flow across it, such as the Godavari, Krishna and Kaveri, which move from west to east. These rivers are very important for farming, drinking water and hydroelectric power. Some rivers, like the Narmada and Tapti, flow westward into the Arabian Sea.

Peninsular Plateau

The Peninsular Plateau is also rich in minerals, forests and fertile soil, making it very important for India's economy. Its forests are home to many tribal communities like the Bhil and Korku, who have their own languages, cultures and live closely connected to nature.

The plateau also has many scenic waterfalls formed where rivers flow over rocky and uneven land. These waterfalls are useful for tourism, irrigation and generating electricity.

INDIA'S AMAZING COASTLINES

India has a long coastline of more than 7,500 kilometres. It is full of natural beauty, with golden sandy beaches, dark rocks, steep cliffs, green forests, coral reefs and dense jungles on some islands.

The West Coast of India

The West Coast of India runs from Gujarat to Kerala, passing through Maharashtra, Goa and Karnataka. Most rivers here flow fast from the Western Ghats, but large rivers like the Narmada and the Tapti flow west into the Arabian Sea, with the Narmada and the Tapti being the biggest. This coast has many important ports and cities that have been centres of trade for thousands of years.

Western Coastal Plains

The East Coast of India

The East Coast lies between the Eastern Ghats and the Bay of Bengal, stretching from the Ganga Delta to Kanyakumari. It has wide plains and large river deltas formed by rivers like the Mahanadi, Godavari, Krishna and Kaveri. These deltas are very fertile and help farmers grow crops. Important water bodies such as Chilika Lake and Pulicat Lake, which are lagoons, are also found along this coast.

Eastern Coastal Plains

Analyze Thoughts

On a blank physical map of India, find and mark the rivers: Godavari, Krishna, and Kaveri.

In-Text Questions

1. Name one tribal community that lives in the forests of the Peninsular Plateau.
2. Which two rivers flow westward into the Arabian Sea from the Western Ghats?
3. Why is Ladakh sometimes called "moonland"?
4. Which famous forts are found in the Aravalli Hills?

INDIAN ISLANDS

India has many islands in the Indian Ocean, Arabian Sea and Bay of Bengal. The two main groups are the Lakshadweep Islands in the Arabian Sea and the Andaman and Nicobar Islands in the Bay of Bengal. These islands are famous for their rare animals, stunning beaches, coral reefs and even volcanoes. Some ancient tribes have lived on these islands for thousands of years.

Lakshadweep Islands

The Lakshadweep Islands are a group of 36 coral islands located near the coast of Kerala. Only a few of these islands have people living on them. This region is important for fishing, studying sea resources and protecting nature.

Coral Reef, Lakshadweep

Andaman and Nicobar Islands

Coral Reef, Andaman and Nicobar

The Andaman and Nicobar Islands are made up of over 500 volcanic islands, divided into two groups—The Andaman Islands in the North and Nicobar Islands in the South. India's only active volcano 'Barren Island', is located here. These islands are important for India's security because they help watch over the surrounding seas. They are home to a wide variety of plants and animals. The Cellular Jail in the Andaman Islands is a famous historical site where freedom fighters were kept during British rule. It reminds us of their brave sacrifices.

In-Text Questions

Write True or False.

1. The Lakshadweep Islands are made up of more than 500 volcanic islands.
2. The Andaman and Nicobar Islands help India monitor the surrounding seas for security purposes.
3. Some ancient tribes have lived on India's islands for thousands of years.

THE SUNDARBANS AND THE DELTA IN WEST BENGAL

Royal Bengal Tiger

If we travel from the islands towards the eastern side of the Himalayas through the Bay of Bengal, we reach the Sundarbans. This area is part of the delta formed by the Ganga, Brahmaputra and their smaller rivers. It is a special place where rivers, land and the sea meet. The Sundarbans is named after the Sundari trees, a type of mangrove that grows well in the salty, muddy water of the region.

Shadsuk Mynsiem Festival

The Sundarbans is shared by two countries—half is in India and half in Bangladesh. It is a protected region and a UNESCO World Heritage Site. Many animals live here, including the famous Royal Bengal Tiger.

THE NORTHEAST HILLS

The north-eastern part of India has a series of high lands that are part of the Meghalaya Plateau. On the map, you can see the Garo, Khasi and Jaintia Hills. This region is famous for its thick forests, heavy rainfall and beautiful waterfalls. The Northeast gets some of the highest rainfall in the world, making the land very fertile and full of plants and animals. Its natural beauty draws many visitors every year.

Indian State of Meghalaya

Go Further

On a blank physical map of India, find and mark the Sundarbans delta. Show the rivers that join to form this delta.

IMPORTANT TERMS

- **Desert:** A dry area with very little rain and few plants.
- **Subcontinent:** A big part of a continent that is clearly separate, like India in Asia.
- **Biodiversity:** The different kinds of plants and animals living in one place.
- **Plateau:** A flat area of land that is higher than the land around it.
- **Delta:** Land formed at the end of a river where it drops sand and soil into the sea.

QUICK TAKE

- India has several major landforms: the Himalayas, the Indo-Gangetic Plains, the Thar Desert, the Peninsular Plateau, the Coastal Plains, and the Islands.
- The Himalayas are the tallest mountains and give rise to many rivers.
- The Indo-Gangetic Plains have fertile soil and support large populations.
- The Thar Desert is dry, with special ways for people and animals to survive and save water.
- The Peninsular Plateau is rich in minerals and home to many tribal communities.
- The coasts and islands are important for trade, wildlife, and cultural connections.

EXERCISES

I. OBJECTIVE TYPE QUESTIONS

A. Choose the correct answer.

1. Which mountain range forms India's northern boundary?
   (a) Western Ghats mountain range ☐ (b) Eastern Ghats hill range ☐
   (c) Himalayan mountain range ☐ (d) Aravalli hill mountain range ☐
2. Why are the Himalayas called the 'Water Tower of Asia'?
   (a) Many rivers originate here ☐ (b) Heavy rainfall every year ☐
   (c) Many lakes are found here ☐ (d) Big glaciers move slowly ☐
3. Which is the highest Himalayan range?
   (a) Shivalik ☐ (b) Himadri ☐
   (c) Himachal ☐ (d) Aravalli hill ☐
4. Which region is known as india's cold desert?
   (a) Thar desert ☐ (b) Ladakh desert ☐
   (c) Rajasthan desert ☐ (d) Kutch desert ☐
5. Which rivers make the Gangetic Plains fertile?
   (a) Narmada and Tapti ☐ (b) Godavari and Krishna ☐
   (c) Ganga and Brahmaputra ☐ (d) Kaveri and Mahanadi ☐

B. Write 'T' True or 'F' for False.

1. India is located in South Asia and is the seventh largest country in the world.
2. The Himalayas form India's northern boundary like a natural wall.
3. The Himalayas are known as the 'Water Tower of Asia'.
4. Ladakh is a hot desert with high rainfall.
5. The Gangetic Plains have fertile soil brought by rivers.
6. The Thar Desert covers parts of Rajasthan and Gujarat.

C. Choose the option that matches items from Column A with Column B.

Column A	Column B
1. Thar Desert	A. Glaciers and snow
2. Himalayas	B. Deltas and Estuaries
3. Coastal Plains	C. Rich in minerals
4. Peninsular Plateau	D. Sand Dunes

(a) 1-C, 2-B, 3-D, 4-A
(b) 1-B, 2-A, 3-D, 4-C
(c) 1-B, 2-A, 3-C, 4-D
(d) 1-D, 2-A, 3-B, 4-C

D. Assertion-Reason

The following questions have two statements – Assertion (A) and Reason (R). Choose the correct answer from the options below:

1. Assertion (A) : The Himalayas are called the Water Tower of Asia.
   Reason (R) : Many major rivers originate from the Himalayas.
2. **Assertion (A)** : The Western Ghats are rich in plants and animals.
   Reason (R) : The Western Ghats are a UNESCO World Heritage Site.
3. **Assertion (A)** : The Aravalli Hills help stop the Thar Desert from spreading eastward.
   Reason (R) : The Aravalli Hills act as a natural barrier.
4. **Assertion (A)** : Ladakh is known as a cold desert.
   Reason (R) : Ladakh receives very little rainfall and has very low temperatures.

(a) Both (A) and (R) are correct, and (R) correctly explains (A).
(b) Both (A) and (R) are correct, but (R) does not explain (A).
(c) (A) is correct, but (R) is wrong.
(d) (A) is wrong, but (R) is correct.

II. PICTURE STUDY

Identify the images and write their names.

III. SHORT ANSWER TYPE QUESTIONS

1. Why is the Sundarbans region important for animals, forests and the people living there?
2. How do mountains, plains and plateaus in India work together to support life and farming?
3. Why are the Eastern Coastal Plains suitable for growing crops and farming activities?
4. How do the Himalayas influence India's rivers and the amount of rainfall in many regions?
5. How do people in the Thar Desert manage their daily life in such a hot and dry climate?

IV. LONG ANSWER TYPE QUESTIONS

1. Explain how rivers that start in the Himalayas help farming, generate hydroelectric power, and support the lives of millions of people in India.
2. Describe how India's different physical features influence transport routes, communication systems and where people choose to live.
3. Imagine the Himalayas stopped rising and began wearing away quickly. How might this change India's climate, rainfall and landforms in the future?

V. SOURCE BASED QUESTIONS

Read the given paragraph and answer the following questions.

The Peninsular Plateau of India

The Peninsular Plateau is one of India's oldest and most important landforms, covering a large part of central and southern India. It is mostly triangular and is called a peninsula because it is surrounded by water on three sides—the Arabian Sea, the Bay of Bengal, and the Indian Ocean. Two main mountain ranges border the plateau. The Western Ghats in the west are tall, continuous and full of forests. During the monsoon, waterfalls flow down their steep slopes. The northern part of this range is called the Sahyadri Hills. The Eastern Ghats in the east are lower, broken, and scattered. Between them lies the Deccan Plateau, a wide flat area. Rivers like the Godavari, Krishna and Kaveri flow eastward, while the Narmada and the Tapti flow west. The plateau is rich in minerals, fertile soil, and forests, supporting tribal communities like the Bhil and Korku. Its waterfalls are useful for tourism, irrigation, and electricity.

1. Name the two mountain ranges that border the plateau.
2. What makes the Western Ghats important during the monsoon?
3. How are the Eastern Ghats different from the Western Ghats?

VI. PROJECT WORK

Exploring a UNESCO Heritage Site

I. Objective:

- Understand the value of India's natural and cultural treasures.
- Improve skills in research, thinking and sharing ideas clearly.

II. Tasks:

- Pick one UNESCO World Heritage place from the chapter (for example, Sundarbans, Great Himalayan National Park, Kumbhalgarh Fort).
- Find details using books, websites or by talking to elders.
- Write a short paragraph describing what makes this site special and why you like it.
- Make a poster or collage with images or sketches and show 3–4 important facts clearly.

III. Expected Learning Outcome:

Students will be able to describe the uniqueness of a UNESCO Heritage Site and express personal reflections.

2 Understanding the Weather

Let's Learn About

- What weather is and what its main elements are such as temperature, rainfall, air pressure, wind, and humidity.
- The work of meteorologists and why weather forecasts are important.
- How traditional knowledge and modern science both help us understand weather.
- Simple and responsible actions to help reduce climate change.

Let's Gear Up

Rahul and Ankit at the Park

Rahul rubbed his hands as a cool wind blew across the park. "Wow, it's really chilly today! I'm glad I wore my jacket," he said. Ankit smiled, "Yes, but think about last month. It was so hot that we needed caps and cold drinks all the time!" Rahul looked up at the sky. "The weather keeps changing. Sometimes it's bright and sunny, sometimes cloudy, windy or even rainy."

Ankit nodded. "These changes—like temperature, rain, wind and humidity—are called the elements of weather. They decide how the day feels."

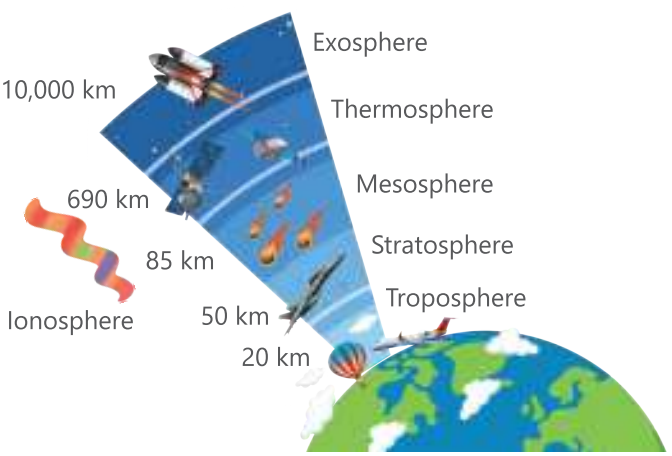
Rahul grinned, "So when we carry umbrellas or wear woollens, we are adjusting to the weather!"

"Exactly," said Ankit. "Let's learn more about how weather works and how it shapes our daily lives."

WEATHER

Weather is how the air around us feels at a particular time and place. But what is the atmosphere? It is the layer of gases that surrounds a planet. On Earth, we call these gases air. You can imagine the atmosphere like a cake with many layers stacked on top of each other.

The layer closest to the Earth's surface is called the troposphere. This is where all land plants, animals and humans live and breathe. It is also where clouds form, rain falls, and winds blow.



Layers of Atmosphere

We describe weather with words like hot, cold, rainy, cloudy, humid, snowy, or windy. Almost all weather events we see—storms, rain and clouds—happen in the troposphere.

This layer stretches about 6 to 18 kilometres above the ground. It is thinner at the poles, where cold air shrinks, and thicker at the equator, where warm air expands.

In-Text Questions

1. The layer of gases closest to the Earth's surface, where clouds form and winds blow, is called the _____.
2. Traditional signs, such as birds flying low or ants carrying eggs, were used by people to predict _____.

ELEMENTS OF WEATHER

Temperature: How hot or cold the air around us is.

Precipitation: Any form of water that falls from the sky, like rain, snow, sleet or hail.

Atmospheric Pressure: The weight of the air above us, pressing down on the Earth's surface.

Wind: The movement of air, including its speed and direction.

Humidity: The amount of water vapour in the air.

Reading the Signals of Nature

Since ancient times, people have been curious about the weather. Long before satellites, radios, or mobile apps existed, they closely watched nature to guess what the weather might bring.

They observed that:

- Birds flying low in the sky often meant rain was coming.
- Ants moving their eggs to higher places warned of heavy rain.
- Frogs croaking loudly in the forests showed that rain might arrive soon.
- Squirrels storing nuts pointed to the arrival of a new season.
- Pine cones opening or closing also gave clues about moisture: they close when it is wet and open when the air is dry.

Over the last few centuries, scientists have developed ways to measure these elements accurately. Using this data, meteorologists predict how the weather will change in a few hours, days or even weeks. Do they just look at the sky and guess? Not at all! They use special instruments to help.

WEATHER INSTRUMENTS

Temperature

Temperature tells us how hot or cold the air, water, or objects are. We measure it using a tool called a thermometer.

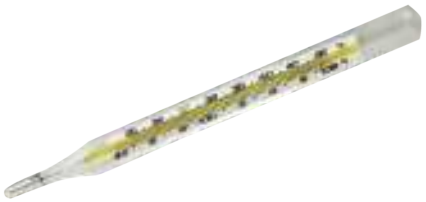
Clinical Thermometer

Types of Thermometers:

Clinical Thermometer: Used to check body temperature.

Laboratory Thermometer: Used to measure the temperature of liquids, chemicals, or other things in experiments.

Temperature can be measured in Celsius (°C) or Fahrenheit (°F). For example, 15°C equals 59°F.



Laboratory Thermometer

Different types of Thermometers

Some thermometers measure the current air temperature, which is called ambient temperature. Others can show the highest (maximum) and lowest (minimum) temperature of the day. For example, in the morning it was 10°C (minimum), and in the afternoon it was 28°C (maximum). These readings can also help calculate important values like:

Range of Temperature: The difference between the maximum and minimum temperatures in 24 hours.

(Range = Maximum – Minimum)

Mean Daily Temperature: The average temperature of the day, calculated by adding the maximum and minimum temperatures and dividing the total by 2.

(Mean = (Maximum + Minimum) ÷ 2)

How do Thermometers Work?

Many thermometers have coloured liquid, usually alcohol or mercury. When it gets hot, the liquid expands and rises. When it cools, the liquid shrinks and drops. Today, we mostly use digital thermometers because they are more accurate, give exact readings and can show highest, lowest and average temperatures.

The India Meteorological Department (IMD) was started in 1875. Its important motto is 'Adityat Jayate Vrishti', which means "From the sun comes rain, from rain comes food, and from food, all living things grow." This line comes from the ancient Indian text Manusmriti and highlights how sun, rain and food are linked to life on Earth.

Go Further

Chart of temperatures of Chandigarh city

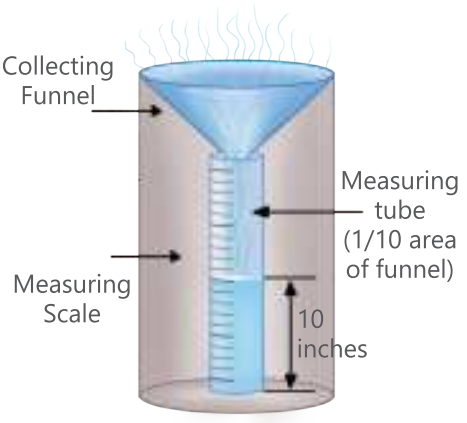
Date	Minimum Temperature (in °C)	Maximum Temperature (in °C)
07.03.2026	15	31
08.03.2026	16	32
09.03.2026	16	32
10.03.2026	16	33
11.03.2026	15	33
12.03.2026	15	32
13.03.2026	15	32

What is the maximum temperature recorded in this week? What is the minimum? Calculate the range for this city?

Precipitation

When the news says that a place received 30 mm of rainfall in one day, it tells us how much rain actually fell there. Imagine keeping a flat plate outside in the open. By the end of the day, the rainwater collected on it would form a layer 30 mm deep.

Rainfall is measured with a tool called a rain gauge. When it rains, water falls into a funnel and collects in a container below. There are markings on the container that show the level of water. If the water level reaches 5 mm, it means the place has received 5 mm of rainfall.



Rain Gauge

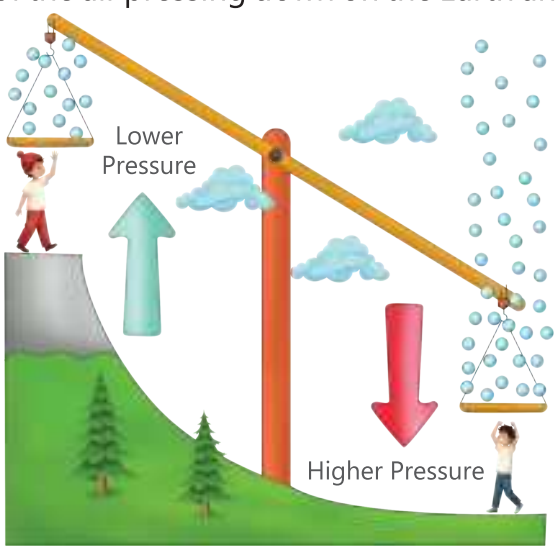
Analyze Thoughts

You can easily make a rain gauge using waste items like plastic bottles, old glass jars, or empty tin cans. Paste a strip of paper or tie a thread outside to mark the scale. Keep it in an open place and check the water level when it rains. Measure and record the rainfall during the next shower.

Atmospheric Pressure

We can easily feel changes in heat and rainfall. But have you noticed that the air feels 'heavy' before a thunderstorm? This happens because of atmospheric pressure. Atmospheric pressure is the weight of the air pressing down on the Earth and everything around us.

Air pressure is higher near sea level and becomes lower as we move up mountains. When you climb higher, the air becomes thin. This means there is less oxygen to breathe. So, people may feel breathless, tired or dizzy at high places because the body must work harder to get enough oxygen.



Atmospheric Pressure

Air pressure is not always high at low levels. Sometimes it drops suddenly. Weather experts call this a depression or a low-pressure system. Such conditions can cause storms or even cyclones. Air pressure is measured with an instrument called a barometer. It is usually measured in millibars (mb). Normal pressure at sea level is about 1013 mb. If it falls below 1000 mb, it shows a depression.

Wind

Wind is moving air. It blows from areas of high air pressure to areas of low air pressure. When we talk about wind, two main things matter:

First, is the direction of the wind (the side it comes from) and second is the speed of the wind (how fast it blows). Wind is very important for our weather in many ways.

- Weather scientists called meteorologists, study wind direction and speed to forecast the weather.

- Pilots and sailors must understand the wind to fly airplanes and sail ships safely.
- Farmers observe the wind to guess where rain clouds might come from.
- Strong winds can also make the soil dry more quickly.

In-Text Questions

1. Why might a mountain climber feel tired and breathless even when the weather is pleasant and sunny?
2. If one city records a maximum temperature of 32°C and a minimum of 20°C, how would you calculate the range of temperature for that day?
3. Why do pilots carefully check wind direction and speed before taking off from an airport?

A wind vane shows the direction of the wind. It has an arrow on one side and a broad tail on the other. When the wind blows, the tail turns, and the arrow points to the direction from which the wind is coming.

Wind Vane

Wind Sock

A windsock is mostly seen at airports. It is a long cloth tube that fills with air and shows wind direction. Pilots look at it to check wind conditions before take-off and landing.

An anemometer measures how fast the wind is blowing. It has small cups attached to arms that rotate when the wind blows. The stronger the wind, the faster the cups turn. The speed is shown on a dial or digital screen in kilometres per hour (km/h).

Anemometer

Humidity

How Do We Measure Humidity?

Humidity is the amount of water vapours present in the air. It is affected by temperature, wind, air pressure and location. For example, places close to the sea usually have more humidity than places far away from it. Humidity tells us how moist or dry the air feels. When humidity is high, we feel warmer and sweat more. When humidity is low, the air feels dry, and water dries up faster. Humidity also influences weather, farming, industries, and even the protection of old objects kept in museums.

Hygrometer

Humidity is measured as relative humidity. Air with no water vapours (which does not happen in nature) is taken as 0%, while air completely filled with water vapours is 100%. Usually, dry weather has relative humidity between 20% and 40%. Humid weather generally ranges from 60% to 80%.

Hygrometer

We measure humidity with an instrument called a hygrometer. It shows the amount of moisture in the air as a percentage. There are different kinds of hygrometers, such as wet and dry bulb hygrometers and digital hygrometers, based on how they measure humidity.

Do you know?

Museums and art galleries use hygrometers to check humidity levels. This helps protect old books, paintings and statues from damage.

WEATHER STATIONS

To understand the weather of a place at any time, we need many instruments. A weather station keeps all these tools together in one area. This helps in measuring and recording weather conditions easily. Weather stations take readings at fixed time gaps. These records help scientists study, map, and forecast the weather.

Automated Weather Station (AWS)

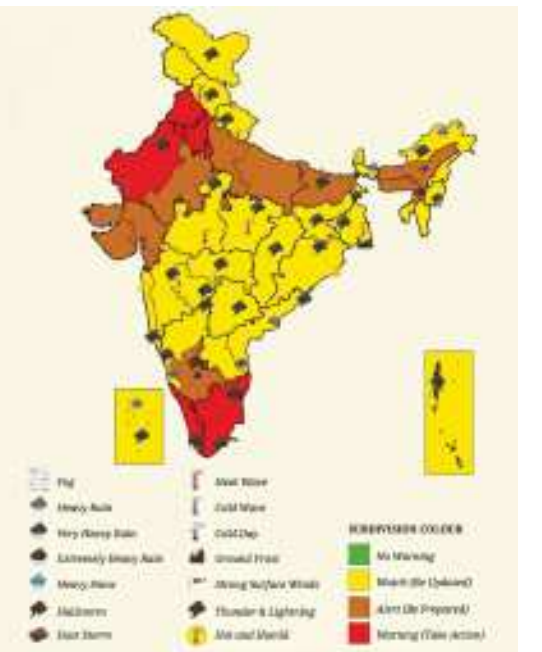
An Automated Weather Station (AWS) works on its own without human control. It uses special sensors to measure and record weather details such as:

- Atmospheric temperature, humidity, wind speed and direction, rainfall and air pressure.
- These stations are useful in farming, air travel, sea travel and checking environmental changes.
- The biggest advantage is that AWS gives correct and quick weather reports automatically.

Automated Weather Station

PREDICTING THE WEATHER

Meteorologists are scientists who study weather. They gather data from different weather instruments over time and carefully examine it. Using scientific methods and computer models, they try to forecast what the weather will be like in the coming days. These predictions are very important today because climate change is causing extreme weather events like droughts, floods, cyclones and storms. Accurate forecasts help people get ready for such conditions and let governments plan ahead. For example, fishermen can be warned not to go to sea during a storm and people in coastal areas can be safely moved before a cyclone hits.



Weather Warning for India: 19 May, 2025

Go Further

In 2023, NDMA installed an AWS at Sikkim's glacial lake. How can it prevent disasters?

IMPORTANT TERMS

- **Atmospheric pressure:** The weight of air on Earth's surface.
- **Humidity:** How much water vapour is in the air.
- **Meteorologist:** Scientist who studies and predicts the weather.
- **Precipitation:** Water falling from the sky, like rain.
- **Temperature:** How hot or cold the air is.
- **Rain gauge:** Tool used to measure rainfall amount accurately.
- **Weather:** How the air feels at a certain place.
- **Atmosphere:** The layer of gases around the Earth.
- **Troposphere:** The lowest layer where weather happens.

QUICK TAKE

- Weather shows the air's condition at a specific place.
- Main weather elements: temperature, precipitation, atmospheric pressure, wind, humidity.
- Weather tools: thermometer (temperature), rain gauge (rain), barometer (pressure), anemometer (wind), hygrometer (humidity).
- Meteorologists use data from weather station and AWS for forecasts.
- Combining traditional knowledge with science improves weather understanding.

EXERCISES

I. OBJECTIVE TYPE QUESTIONS

A. Choose the correct answer.

1. What is weather?
   (a) Air around us ☐ (b) Sunlight only ☐
   (c) Clouds only ☐ (d) Rain only ☐
2. Rainfall is measured by?
   (a) Thermometer ☐ (b) Rain gauge ☐
   (c) Windsock ☐ (d) Barometer ☐
3. Humidity is measured by?
   (a) Barometer ☐ (b) Thermometer ☐
   (c) Rain gauge ☐ (d) Hygrometer ☐
4. Air pressure is measured by?
   (a) Thermometer ☐ (b) Barometer ☐
   (c) Windsock ☐ (d) Hygrometer ☐
5. Wind speed is measured by?
   (a) Thermometer ☐ (b) Wind vane ☐
   (c) Anemometer ☐ (d) Hygrometer ☐

B. Write 'T' for True or 'F' for False.

1. The troposphere is thinner at the equator.
2. Precipitation includes snow and hail.
3. Barometers measure wind speed.
4. Windsocks show wind direction.
5. Digital thermometers cannot show maximum temperature.

C. Fill in the blanks.

1. The layer of gases that surrounds the Earth is called the _____.
2. Almost all weather events happen in the _____ layer of the atmosphere.
3. The tool used to measure rainfall is called a _____.
4. Air moves from areas of high pressure to areas of low pressure, and this movement is called _____.
5. The average temperature of the day can be calculated using the maximum and minimum temperatures and is called the _____.

D. Assertion-Reason

The following questions consist of two statements - Assertion (A) and Reason (R). Answer these questions by selecting the correct option:

1. **Assertion (A)** : Wind always blows from areas of low pressure to areas of high pressure.
   Reason (R) : Wind is caused by differences in atmospheric pressure.
2. **Assertion (A)** : Meteorologists use instruments and computer models to predict the weather.
   Reason (R) : Observing clouds alone is sufficient to forecast weather accurately.
3. **Assertion (A)** : A rain gauge measures the amount of rainfall.
   Reason (R) : Rainwater collected in a funnel and container can be measured in millimeters.
4. **Assertion (A)** : Atmospheric pressure is the weight of the air pressing down on Earth.
   Reason (R) : Air pressure decreases with altitude, making it harder to breathe at high places.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false and (R) is true.

II. PICTURE STUDY

Look at the images carefully. Can you identify the weather instrument shown in each picture? Write their names.

_____

_____

_____

III. SHORT ANSWER TYPE QUESTIONS

1. How can old traditional weather signs, like birds and pine cones, help scientists make better forecasts today?
2. How did people in ancient times connect the sun with the timing of rain and monsoon?
3. How does an Automated Weather Station (AWS) help communities prepare for changing climate conditions?

IV. LONG ANSWER TYPE QUESTIONS

1. Why is learning about weather and meteorology useful in our daily life, from farming to traveling safely?
2. What is atmospheric pressure? How does it affect us in higher altitudes?
3. Explain why different weather instruments, like thermometers, barometers and rain gauges, are all important for forecasting weather.

V. SOURCE BASED QUESTIONS

Read the given paragraph and answer the following questions.

Weather Forecasting for Food Security

In a drought-prone village in Maharashtra, farmers often lost crops because they relied only on traditional signs, like birds or animal behaviour, to guess rainfall. Recently, the village started receiving IMD weather updates on mobile phones. Farmers also set up a simple rain gauge and barometer at the community centre. That year, they waited to sow seeds until the IMD confirmed enough rainfall. They also used forecasts to plan irrigation during dry periods. By combining traditional knowledge with scientific forecasts, the farmers harvested more grain than in earlier years.

(a) What did the rain gauge measure at the centre?

(b) Why did farmers lose crops before using weather forecasts?

(c) How can mobile phones help share IMD warnings quickly in remote areas?

VI. PROJECT WORK

"Rainfall in Millimeters" – From Drops to Millimeters

I. Objective:

- Understand rainfall depth (mm) and why it's better than "litres" for comparing places.

II. Tasks:

- Using the DIY rain gauge, collect 3-5 days of data.
- Convert & visualize: Make a simple bar chart (day vs. mm).
- Apply: If 25 mm falls on the school ground (e.g., 2,000 m^2), how deep is the water layer? What would that mean for runoff, puddles, or garden watering?
- Discuss: Why farmers/planners care about mm, intensity, and distribution (drought/flood risk, sowing windows).

III. Expected Learning Outcome:

- Interpret mm rainfall as a depth applicable anywhere.
- Link rainfall data to real decisions (water use, drainage).